Linear Algebra and Calculus Lecture 7

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Linear Independence

Definition. A set of vectors $\{v_1, \ldots, v_p\}$ in $\mathbb{R}^n$ is called **linearly independent** if the vector equation

$$x_1v_1 + x_2v_2 + \cdots + x_pv_p = 0$$

only has solution x = 0. It is called **linearly dependent** if the previous vector equation has nontrivial solutions.

Proposition. *If A is an $m \times n$ matrix with columns $a_1, \ldots, a_n \in \mathbb{R}^m$, then the vectors $a_1, \ldots, a_n$ are linearly independent if and only if the matrix equation $Ax = 0$ has no nontrivial solution.*

Question 1. *Let $v \in \mathbb{R}^n$. Is v necessarily linearly independent?*

Exercise. Determine if the following sets of vectors are linearly independent.

- $v_1 = \left[\begin{smallmatrix} 3 \\ 1 \end{smallmatrix}\right], v_2 = \left[\begin{smallmatrix} 6 \\ 2 \end{smallmatrix}\right]$.
- $v_1 = \left[\begin{smallmatrix} 3 \\ 2 \end{smallmatrix}\right], v_2 = \left[\begin{smallmatrix} 6 \\ 2 \end{smallmatrix}\right]$.

Proposition. *Two non-zero vectors v_1, v_2 are linearly independent if and only if v_1 is not a scalar multiple of v_2.*

Theorem. *(Characterization of Linearly Independent Sets) An indexed set $S = \{v_1, \ldots, v_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others. In fact, if S is linearly dependent and $v_1 \neq 0$, then some v_j (with $j > 1$) is a linear combination of the preceding vectors, $v_1, \ldots, v_{j-1}$.*

Theorem. *If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set $\{v_1, \ldots, v_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.*

Example. Any matrix with more columns than rows will have linearly dependent columns.

Theorem. *If a set $S = \{v_1, \ldots, v_p\}$ in $\mathbb{R}^n$ contains the zero vector, then the set is linearly dependent.*

Linear Transformations

Definition. A **transformation** (or **map** or **function**) $T : \mathbb{R}^n \to \mathbb{R}^m$ is a rule that assigns to each vector $x \in \mathbb{R}^n$ a vector $T(x) \in \mathbb{R}^m$.

The definitions of domain, codomain, range, image, etc are the same as in calculus.
In this section we will focus on *matrix maps*, that is, linear maps T in which $T(x) = Ax$ for some matrix A of size $m \times n$.

Example. Let

$$A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}, c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}.$$

Let's determine if c is in the range of the map T given by $T(x) = Ax$.

The vector c is in the range of T if c is the image of some $x \in \mathbb{R}^2$, that is, if $c = T(x)$ for some x. This is just another way of asking if the system $Ax = c$ is consistent. To find the answer, row reduce the augmented matrix:

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 14 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{bmatrix}$$

The third equation, $0 = -35$, shows that the system is inconsistent. So c is not in the range of T.

Definition. A transformation (or mapping) T is **linear** if

1. $T(u + v) = T(u) + T(v)$, for all u, v in the domain of T;
2. $T(cu) = cT(u)$ for all scalars c and all u in the domain of T.

Remark. Every matrix transformation is a linear transformation (since $A(u + v) = Au + Av$ and $A(cu) = cAu$), but not every linear transformation is a matrix transformation.

Remark. If T is a linear transformation, then $T(0) = 0$, and $T(cu + dv) = cT(u) + dT(v)$, for all vectors u, v in the domain of T and all scalars c, d.

Of course, this can be generalized to

$$T(c_1v_1 + \cdots c_pv_p) = c_1T(v_1) + \cdots c_pT(v_p).$$

In fact, this last conditions summarizes the two conditions of the definition, so it is enough to prove linearity.

Example. Given a scalar r, define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(x) = rx$. T is called a **contraction** when $0 \leq r \leq 1$ and a **dilation** $r > 1$. Let's prove it is a linear transformation. Let $u, v \in R^2$ and let c, d be scalars. Then

$$T(cu + dv) = r(cu + dv) = r(cu) + r(dv) = c(ru) + d(rv) = cT(u) + dT(v).$$

Exercise. Let T be a matrix map given by $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Find the images under T of $u = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $u + v = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$.